PYTHON FOR BEGINNERS AND PROFESSIONALS

SESSION 04

Loops – WHILE and FOR loop

What is a Loop?

A loop lets you run the same block of code multiple times without writing it again and again. Instead of doing this:

```
print("Hello")
print("Hello")
print("Hello")
print("Hello")
print("Hello")
```

You do this:

```
i = 0
while i < 5:
    print("Hello")
    i += 1
```

Types of Loops

There are two types of loops that we will learn:

1. WHILE loop
2. FOR loop

WHILE Loop

A while loop keeps running as long as its condition is True. Python checks the condition before every iteration – the moment it becomes False, the loop stops.

```
while condition:
    # this block runs repeatedly until condition is False
```

```
count = 1

while count <= 5:
    print(f"Count is {count}")
    count += 1

# Output:
# Count is 1
# Count is 2
# Count is 3
# Count is 4
# Count is 5
```

Practice Questions on WHILE Loop

Q15. Print all the numbers which are divisible by 3 and 5, from 1 to 100.

Q16. Sum of all the numbers from 1 to 100.

Q17. Sum of all the numbers from 1 to 100 divisible by 2 and 7.

Q18. Ask a number from the user, print the multiplication table upto 10.

Q19. Ask a number from the user, and print all the factors.

FOR loop

A ***for*** loop is used to iterate over a sequence – a range of numbers, a string, a list, and so on. Unlike while, you don't manage a counter manually – Python handles it for you.

```
for variable in sequence:
    # this block runs for each item in the sequence
```

```
# Looping over a range of numbers
for i in range(1, 6):
    print(i)
# Output: 1 2 3 4 5
```

BREAK statement

break immediately stops the loop and exits it, even if the condition is still True or there are items left in the sequence.

```
num = 1
while num <= 10:
    if num == 5:
        break          # stops the loop when num reaches 5
    print(num)
    num += 1
# Output: 1 2 3 4

# break in a for loop
for i in range(1, 11):
    if i == 6:
        break
    print(i)
# Output: 1 2 3 4 5
```

CONTINUE statement

continue skips the rest of the current iteration and jumps straight to the next one. The loop does not stop – it just skips that particular cycle.

```
# continue in a for loop – skip even numbers
for i in range(1, 11):
    if i % 2 == 0:
        continue      # skip even numbers
    print(i)
# Output: 1 3 5 7 9

# continue in a while loop – skip multiples of 3
num = 0
while num < 15:
    num += 1
    if num % 3 == 0:
        continue
    print(num)
# Output: 1 2 4 5 7 8 10 11 13 14
```

Practice Question on BREAK and CONTINUE

Q20. Take numbers as input from the user one by one. Skip negative numbers and keep adding the positive ones. Stop when the user enters 0 and print the total. (Uses both continue and break.)

Nested Loops

A nested loop is a loop inside another loop. The inner loop completes all its iterations for every single iteration of the outer loop.

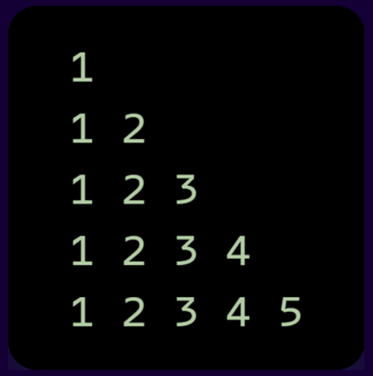
```python
for outer in range(1, 4):
    for inner in range(1, 4):
        print(f"outer={outer}, inner={inner}")

# Output:
# outer=1, inner=1
# outer=1, inner=2
# outer=1, inner=3
# outer=2, inner=1
# outer=2, inner=2
# outer=2, inner=3
# outer=3, inner=1
# outer=3, inner=2
# outer=3, inner=3
```

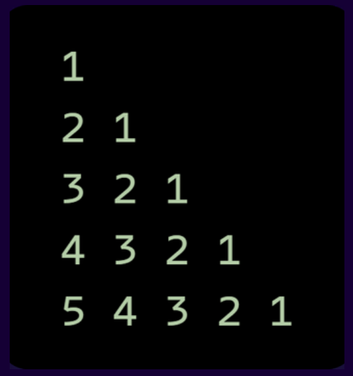
Practice Questions for Nested Loops

Q21.

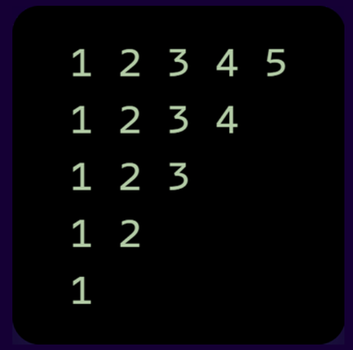
Q22.



Q23.

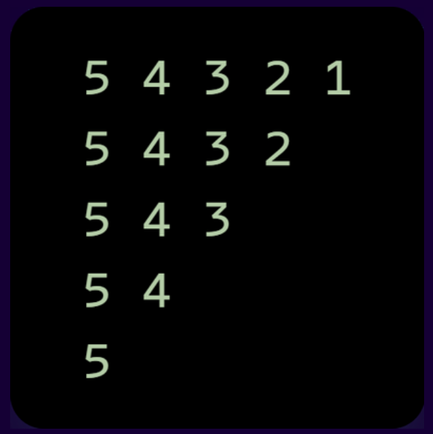


Q24.

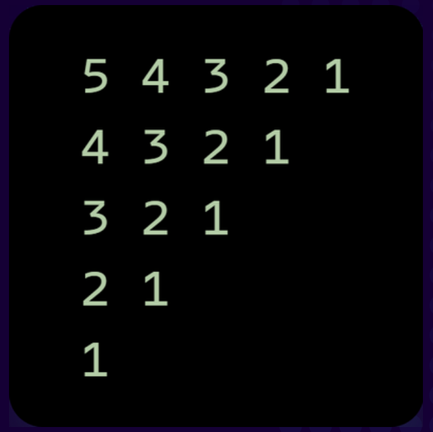


Practice Questions for Nested Loops

Q25.

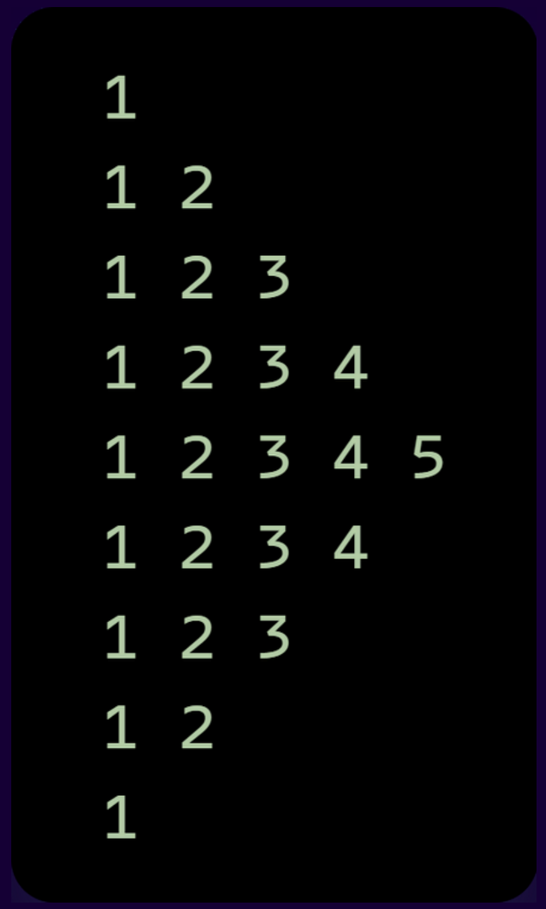


Q26.

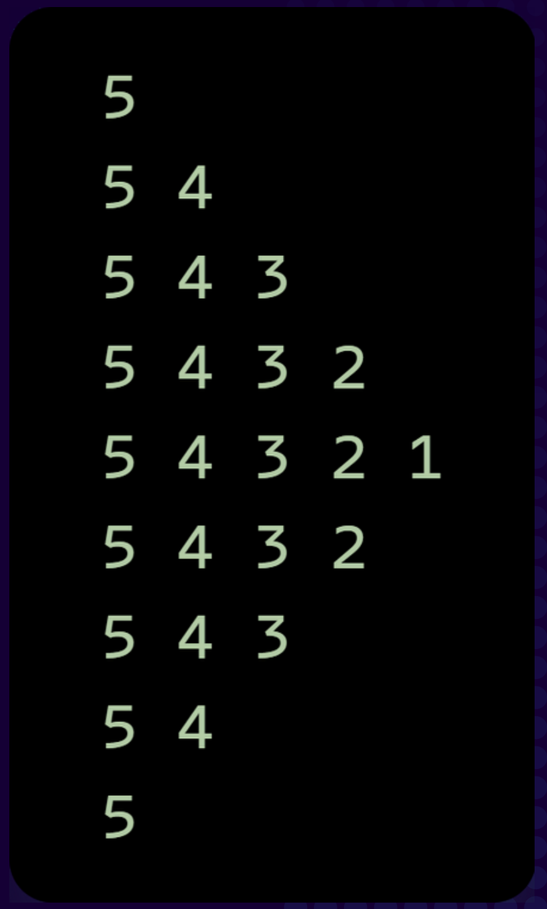


Practice Questions for Nested Loops

Q27.

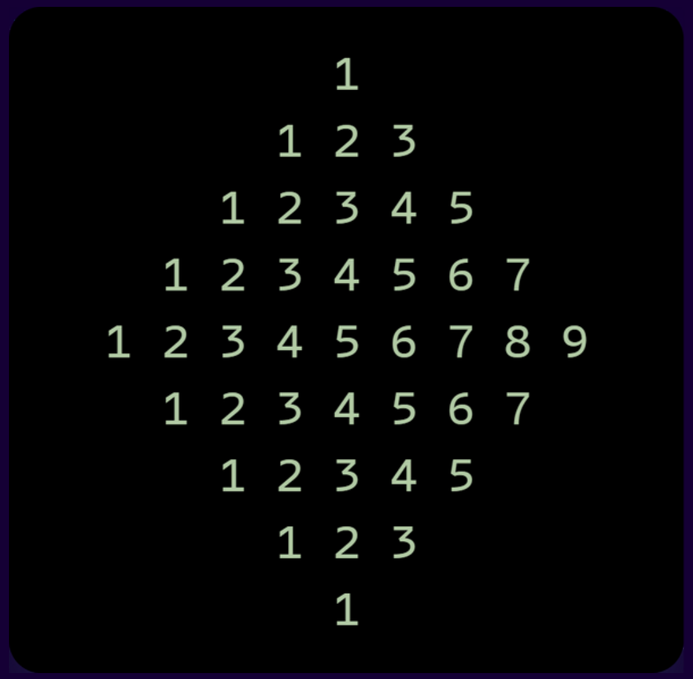


Q28.

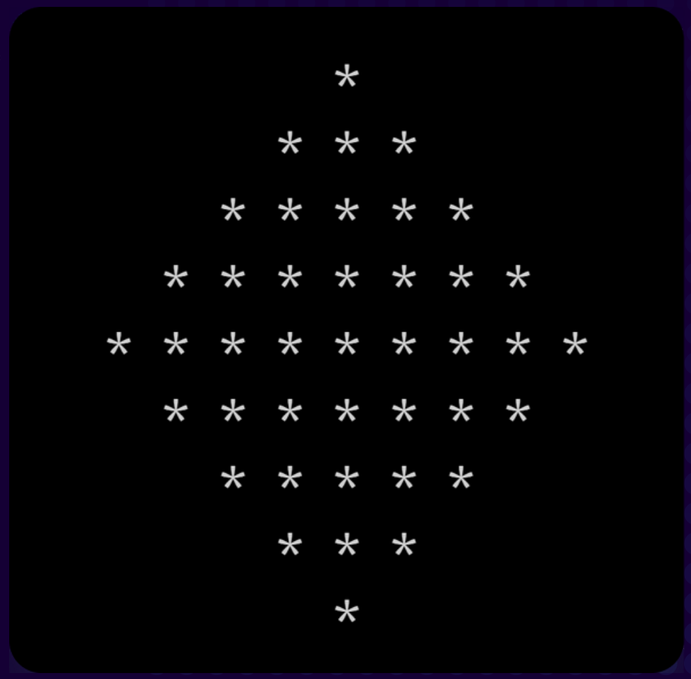


Practice Questions for Nested Loops

Q29.

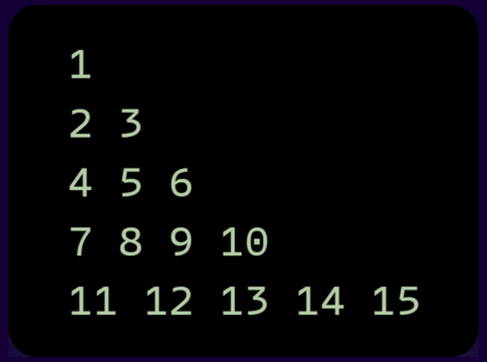


Q30.



Practice Questions for Nested Loops (Assignment)

Q31.



Q32.

Q33.

Q34.